SMARTBRIDGE

LET'S BRIDGE THE GAP

SKILL WALLET

A SmartBridge Product

Date	7th July 2026
Team ID	XXXXXX
Project Name	edugenie google gemini powered learning assistant
Maximum Marks	5 Marks

CODING & SOLUTION

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

SOLUTION SUMMARY

FIELD	DETAILS
Repository Link / URL	https://github.com/team-XXXXXX/edugenie-gemini-assistant
Programming Language(s)	Python, JavaScript (TypeScript)
Framework(s) Used	FastAPI (Backend), React.js & Tailwind CSS (Frontend), Google GenAI SDK

FIELD	DETAILS
Key Features Implemented	• **AI-Powered Doubt Resolution:** Contextual real-time answering using Gemini 1.5 Flash. • **Smart Quiz Generator:** Generates dynamic conceptual tests based on uploaded notes/syllabi. • **Automated Progress Tracker:** Uses customized roadmaps to guide student milestones. • **Interactive Document Summarizer:** Processes textbooks/PDFs to provide instant TL;DR and key takeaways.
Pending/Incomplete Features	• Voice-to-Text interaction module for accessibility. • Offline local storage synchronization for study progress tracking.
Setup / Run Instructions	**Backend Setup:** 1. Navigate to `/backend`, run `pip install -r requirements.txt`. 2. Configure `.env` with `GEMINI_API_KEY`. 3. Launch server via `uvicorn main:app --reload`. **Frontend Setup:** 1. Navigate to `/frontend`, run `npm install`. 2. Start localized dashboard using `npm start`.

CODE QUALITY CHECKLIST

S.NO	CRITERIA	STATUS (YES/ NO)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

ADDITIONAL NOTES / COMMENTS:

The application backend features robust exception handling for the Google Gemini API integration to gracefully handle token rate limits and potential network delays. The frontend codebase is split into reusable React hooks for API interaction, ensuring strict separation of concerns and high readability.